ETH zürich



Introduction to Scientific Computation
Lecture 6
Fall 2026

Graphs

Mikhail Usvyatsov, Yury Kartynnik October 28, 2026

Graph

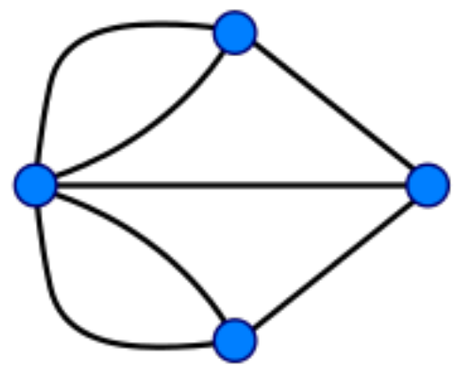
Seven Bridges of Königsberg











Leonhard Euler, 1736

Graph

$$G :< V, E >$$



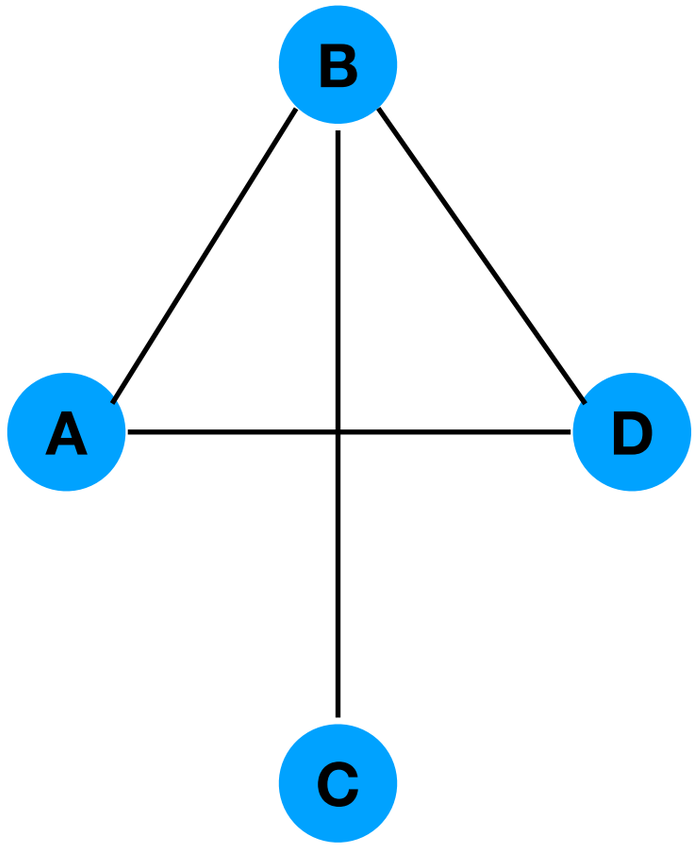
V node in the graph

E edge in the graph, the connection between two nodes

Applications

- Road networks
- Electronic circuits
- Telecommunication networks
- Social networks
- Any relationships …

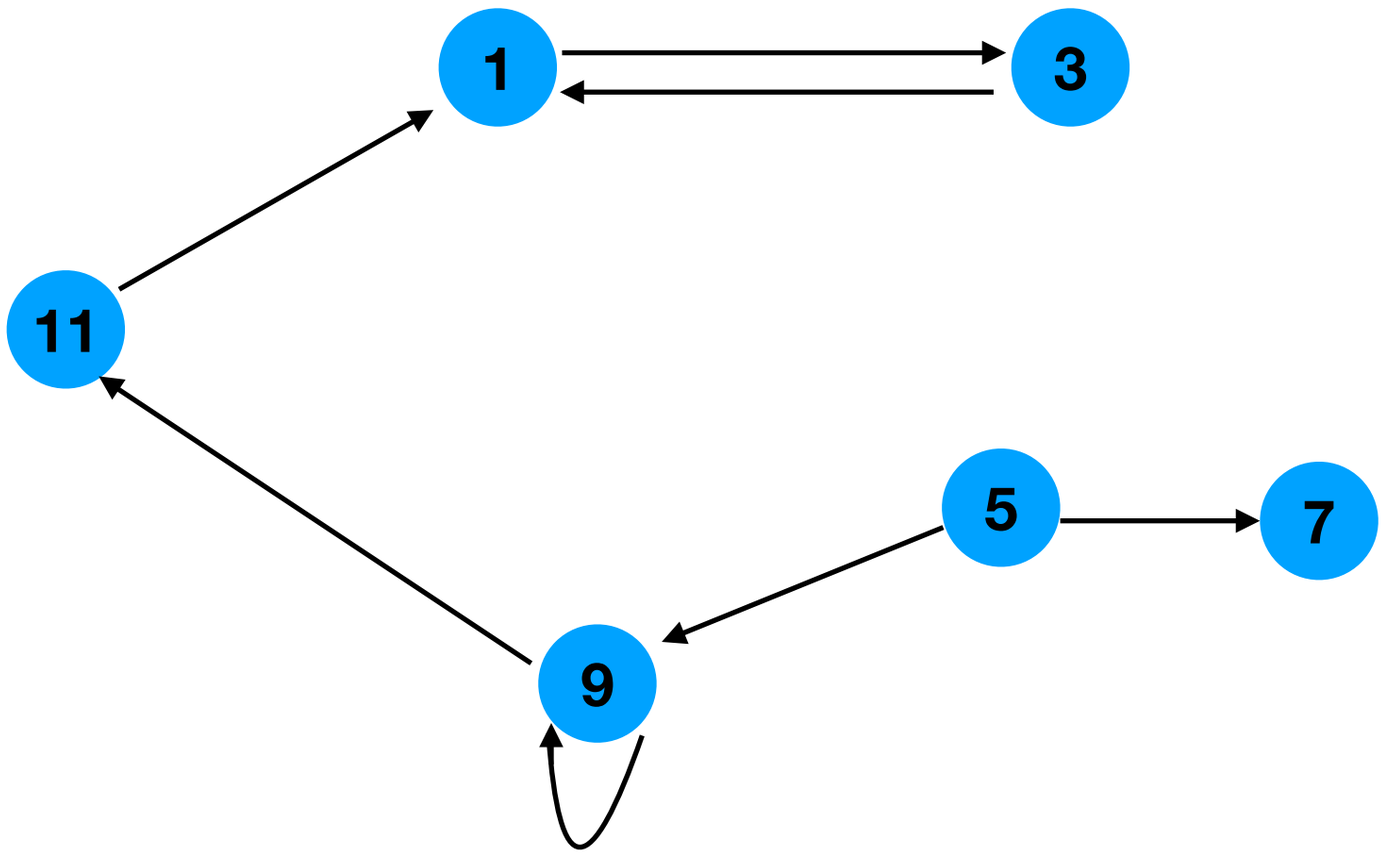
Types



Undirected G

$$V = \{A, B, C, D\}$$
$$E = \{AB, BD, AD, BC\}$$

Types



Directed G

$$V = \{1,3,5,7,9,11\}$$

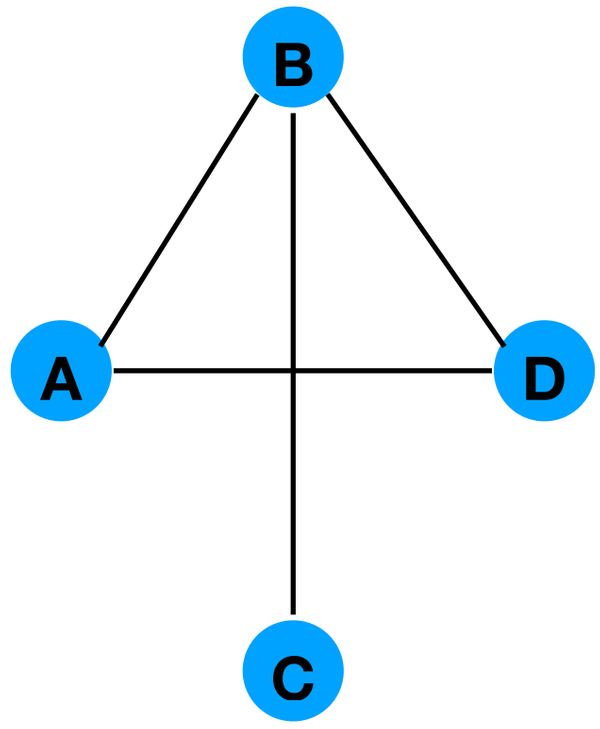
$$E = \{(1,3),(3,1),(5,7),(5,9),(5,7),(5,9),(9,9),(9,11),(11,1)\}$$

Definitions

Adjacency - two vertices are called **adjacent** if they are connected by edge

Path - the sequence of vertices which connects two nodes in a graph

Complete graph - every vertex is connected to every other vertex



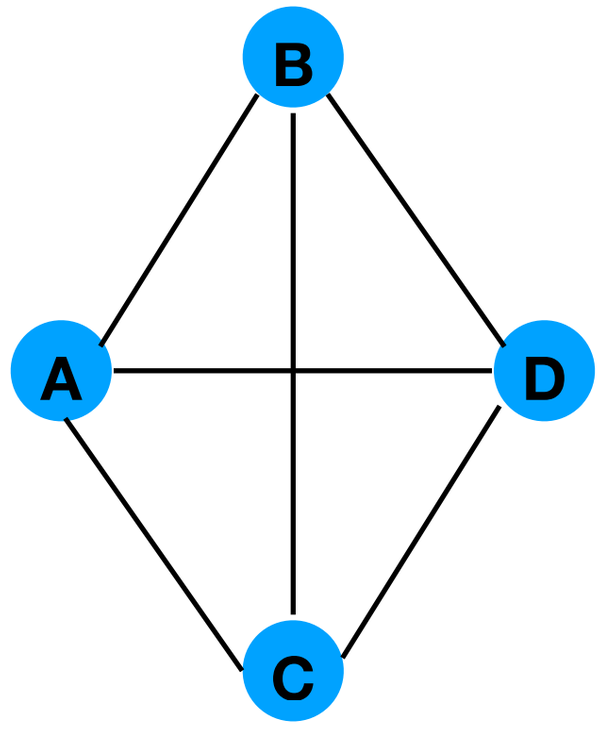
Wighted graph - graph with the values assigned to edges

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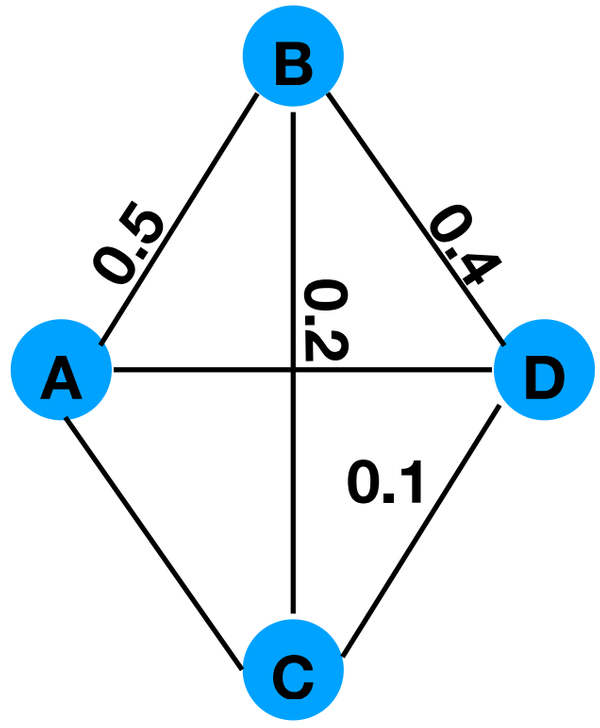
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Definitions

Quiz: how many edges exist in a complete graph ?

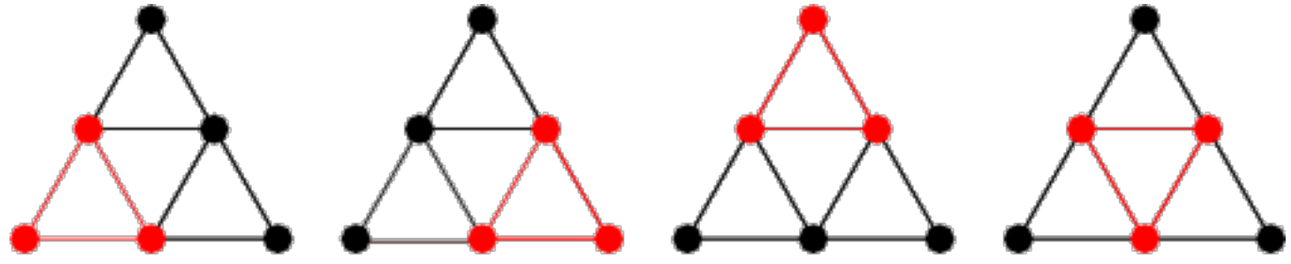
Definitions

Quiz: how many edges exist in a complete graph ?

Answer: $\frac{N^2 - N}{2}$

Definitions

Clique - complete subgraph



cliques

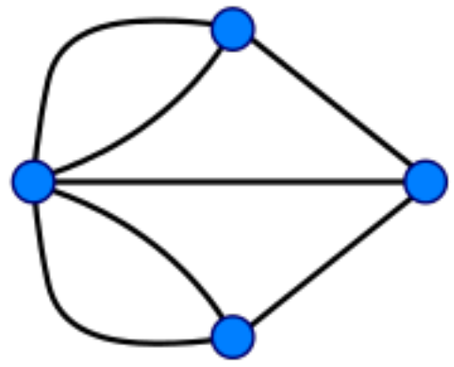
Euler trail (path) - the path in a finite graph which visits every edge exactly once

Degree $d(V)$ **-** number of edges incident to V

Definitions

Clique - complete subgraph

Euler trail (path) - the path in a finite graph which visits every edge exactly once



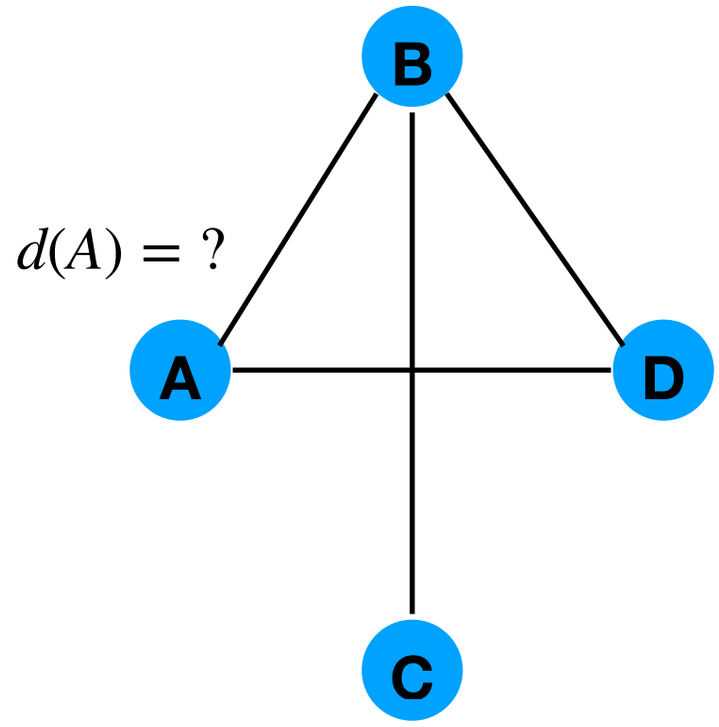
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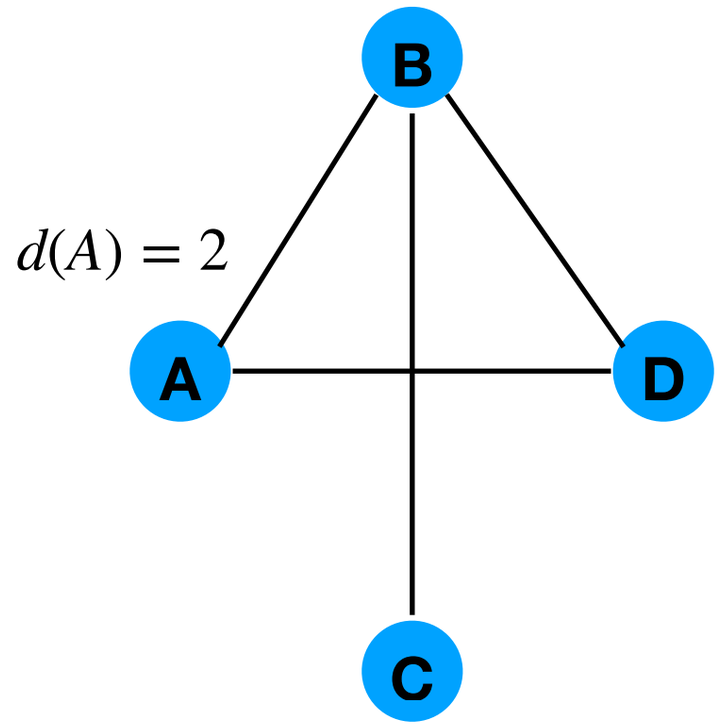


Definitions

Clique - complete subgraph

Euler trail (path) - the path in a finite graph which visits every edge exactly once

Degree $d(V)$ **-** number of edges incident to V



Definitions

Handshaking lemma

$$G = < V, E >$$

$$\sum_{u \in V} d(u) = 2\,|E|$$

in a party of people some of whom shake hands, an even number of people must have shaken an odd number of other people's hands.

Graph

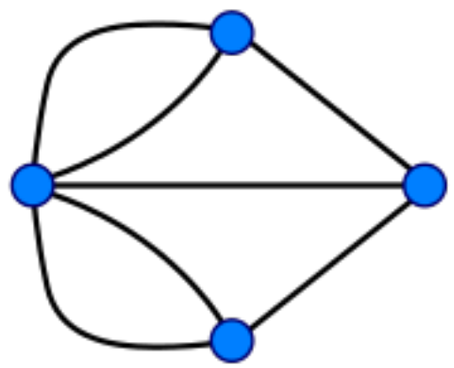
Back to Seven Bridges of Königsberg







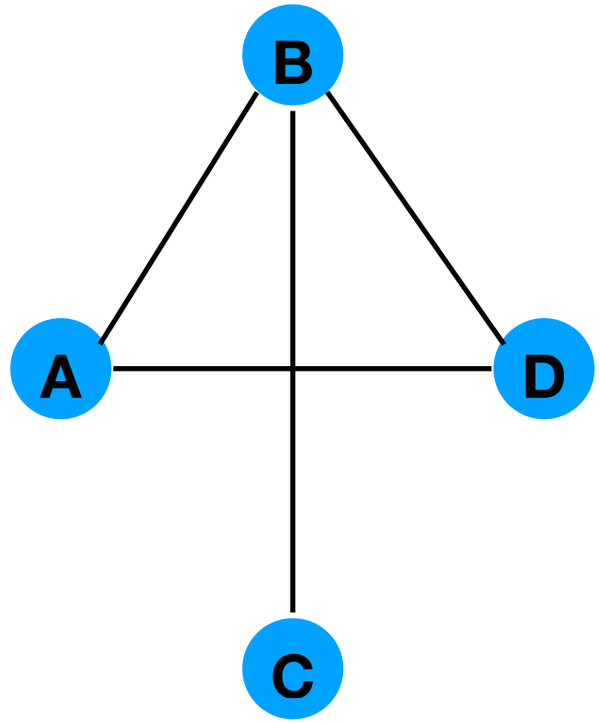




An undirected graph has an Euler path if and only if exactly zero or two vertices have odd degree, and all of its vertices with nonzero degree belong to a single connected component.

Leonhard Euler, 1736

CS representation



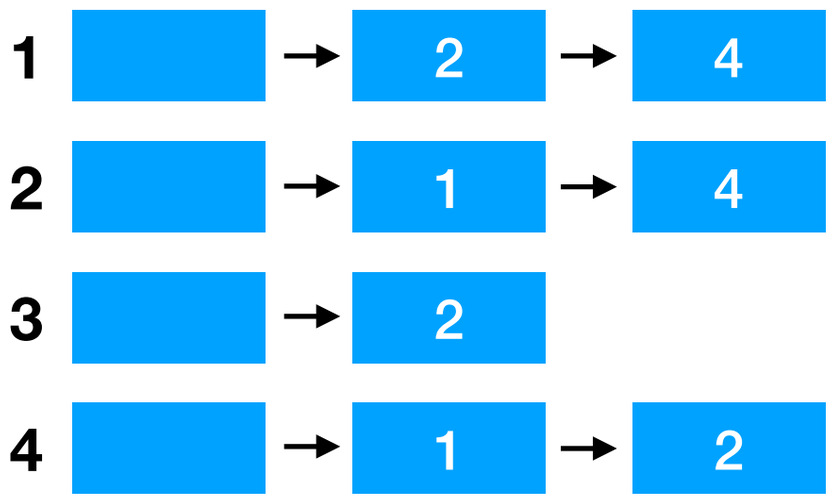
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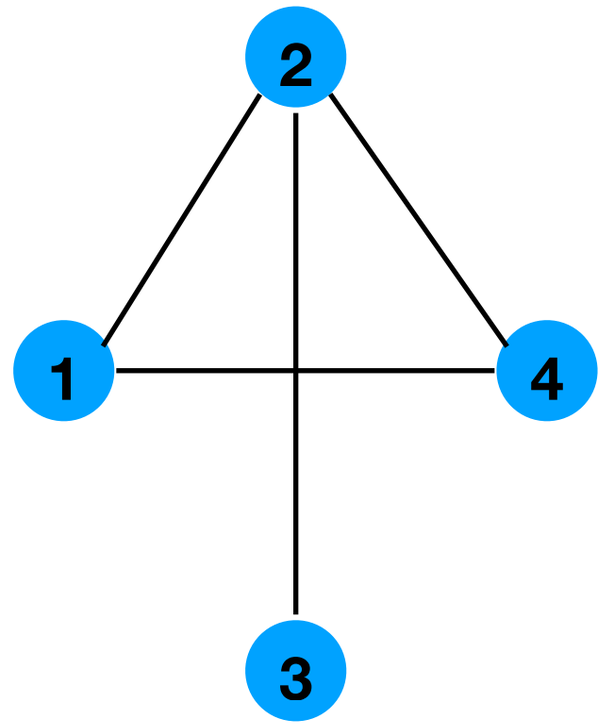
$$G = < V, E >$$

Adjacency matrix

	1	2	3	4
1	0	1	0	1
2	1	0	1	1
3	0	1	0	0
4	1	1	0	0

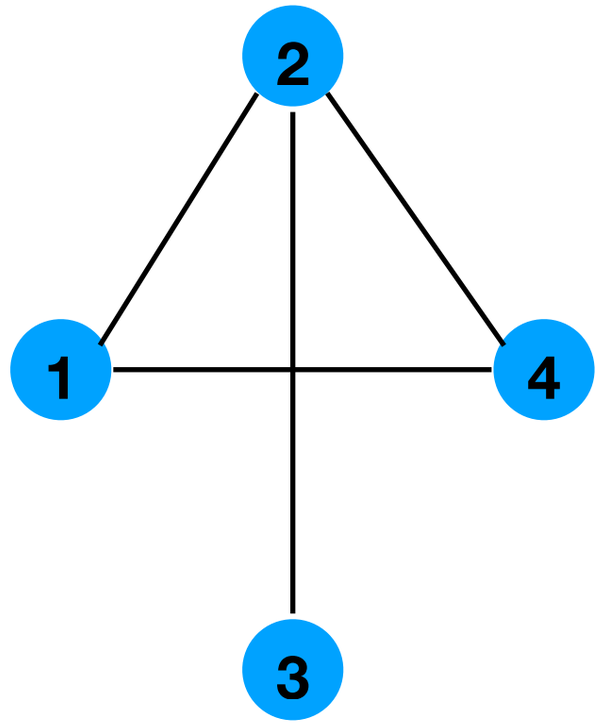
Adjacency list





CS representation

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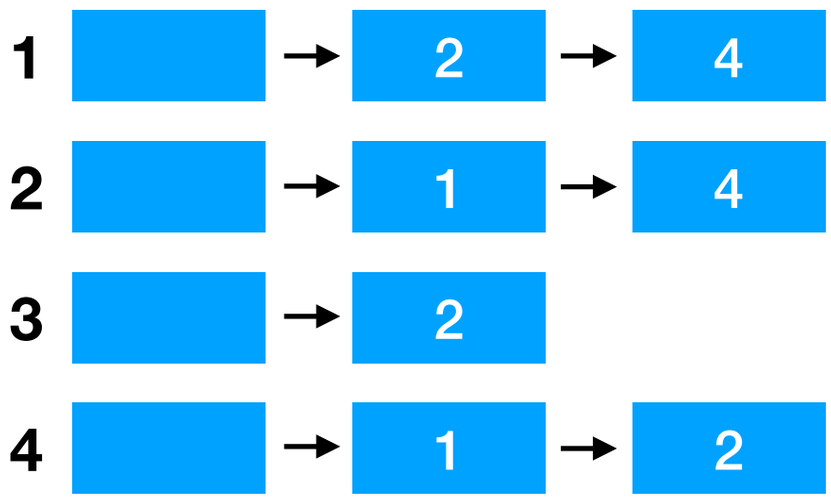


Adjacency matrix

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Directed: N^2
Undirected: N^2

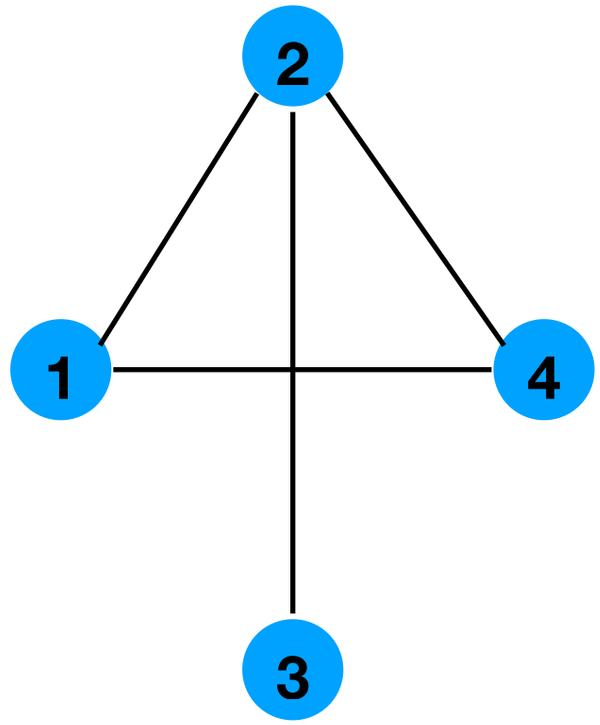
Adjacency list



Directed: $N + M$
Undirected: $N + 2M$

CS representation

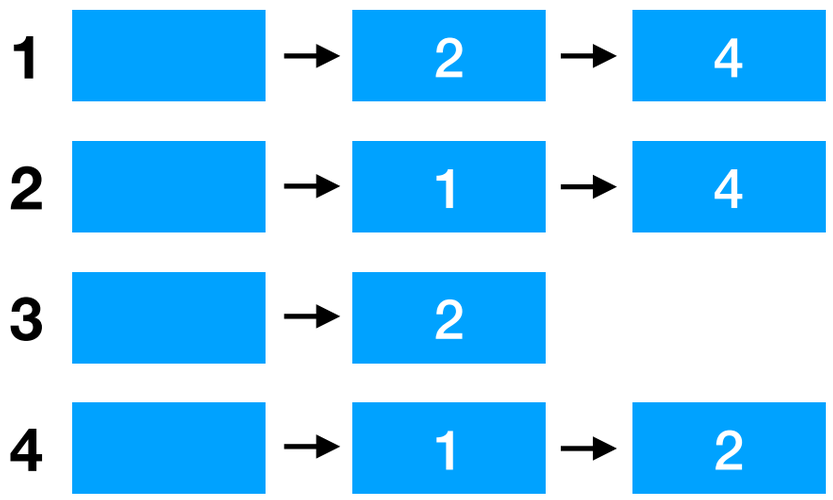
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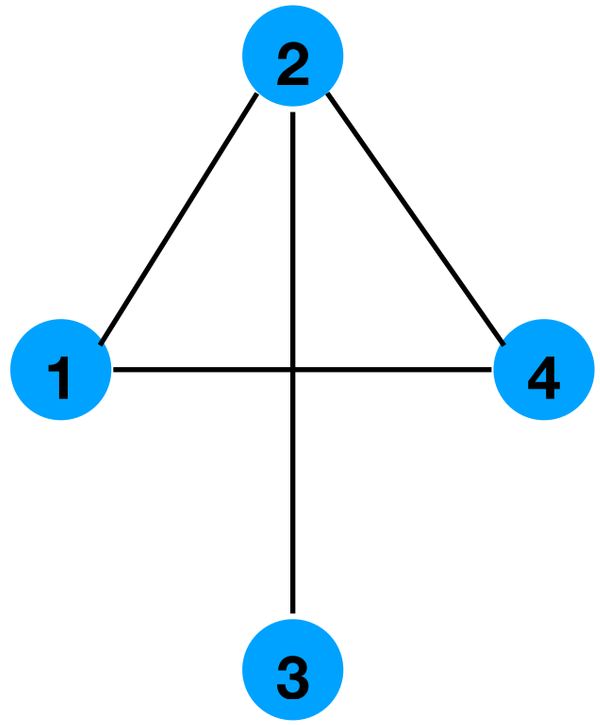
Adjacency list



Quiz: What is better to test if an edge is in the graph ?

CS representation

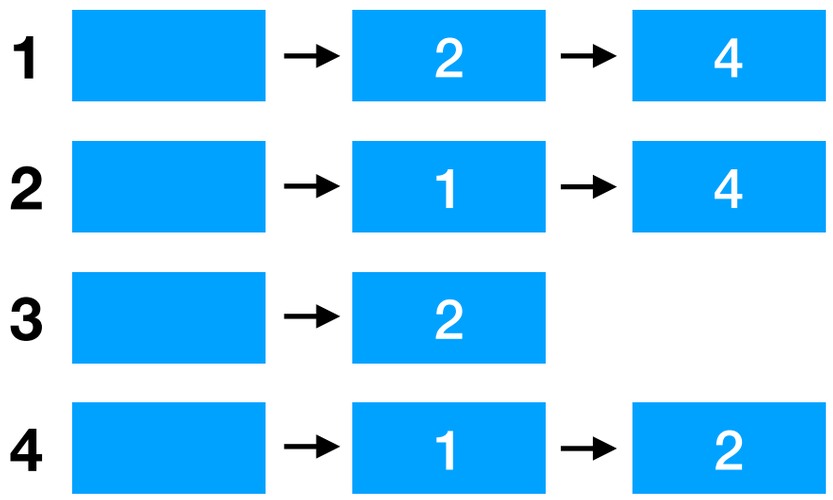
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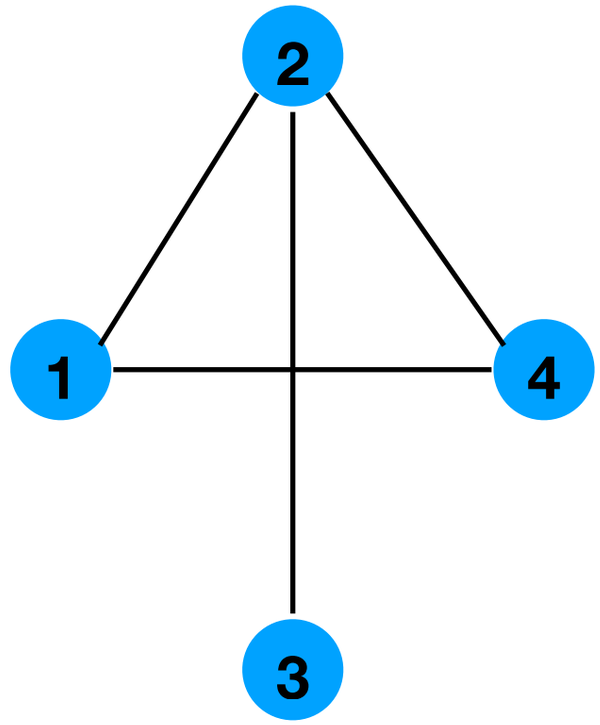


Quiz: What is better to test if an edge is in the graph ?

Answer: matrix

CS representation

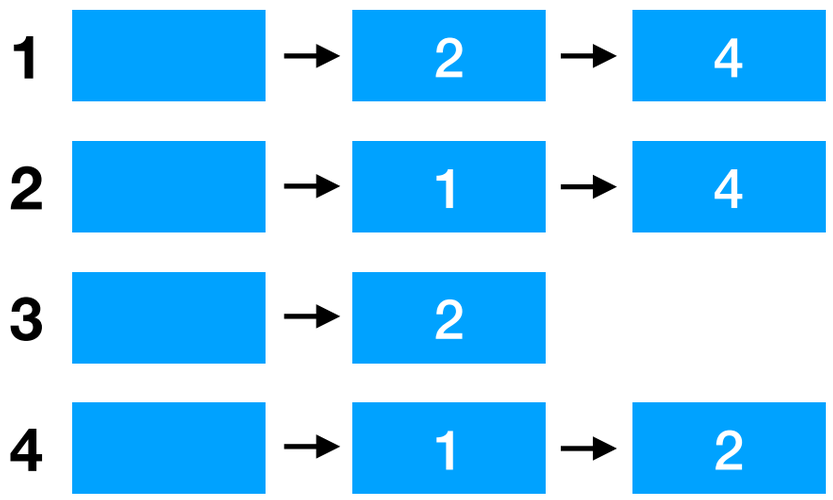
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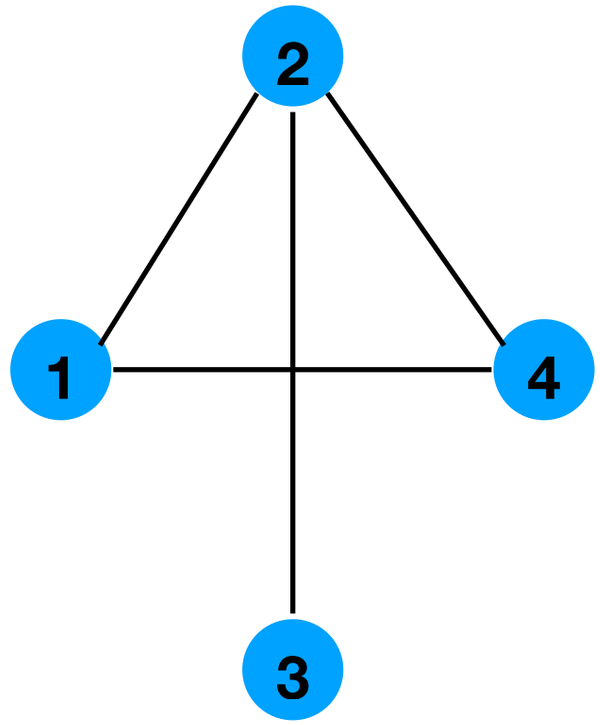
Adjacency list



Quiz: What is faster to find the degree of vertex ?

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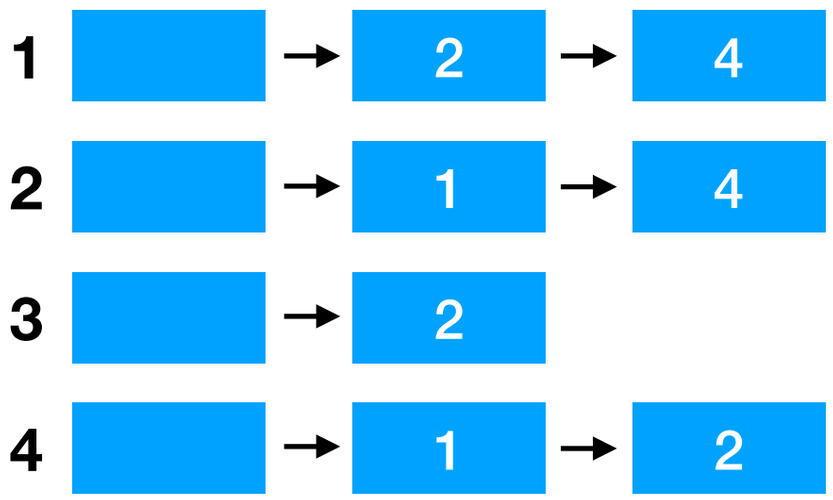
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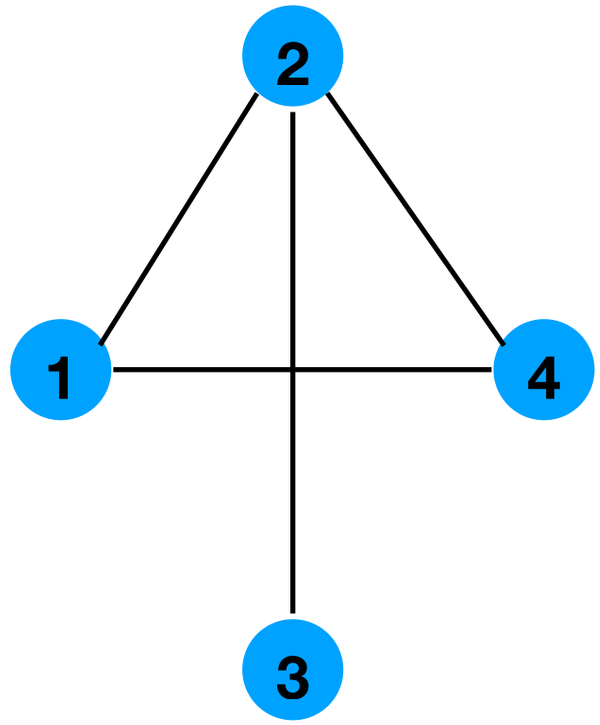


Quiz: What is faster to find the degree of vertex ?

Answer: list

CS representation

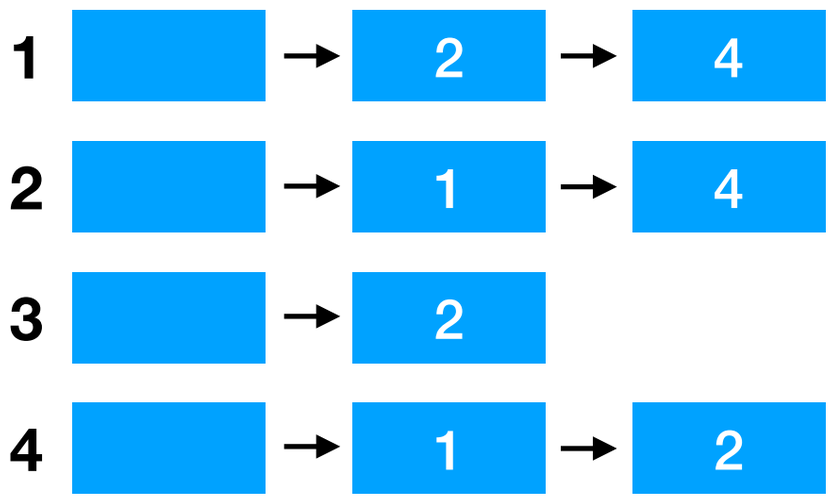
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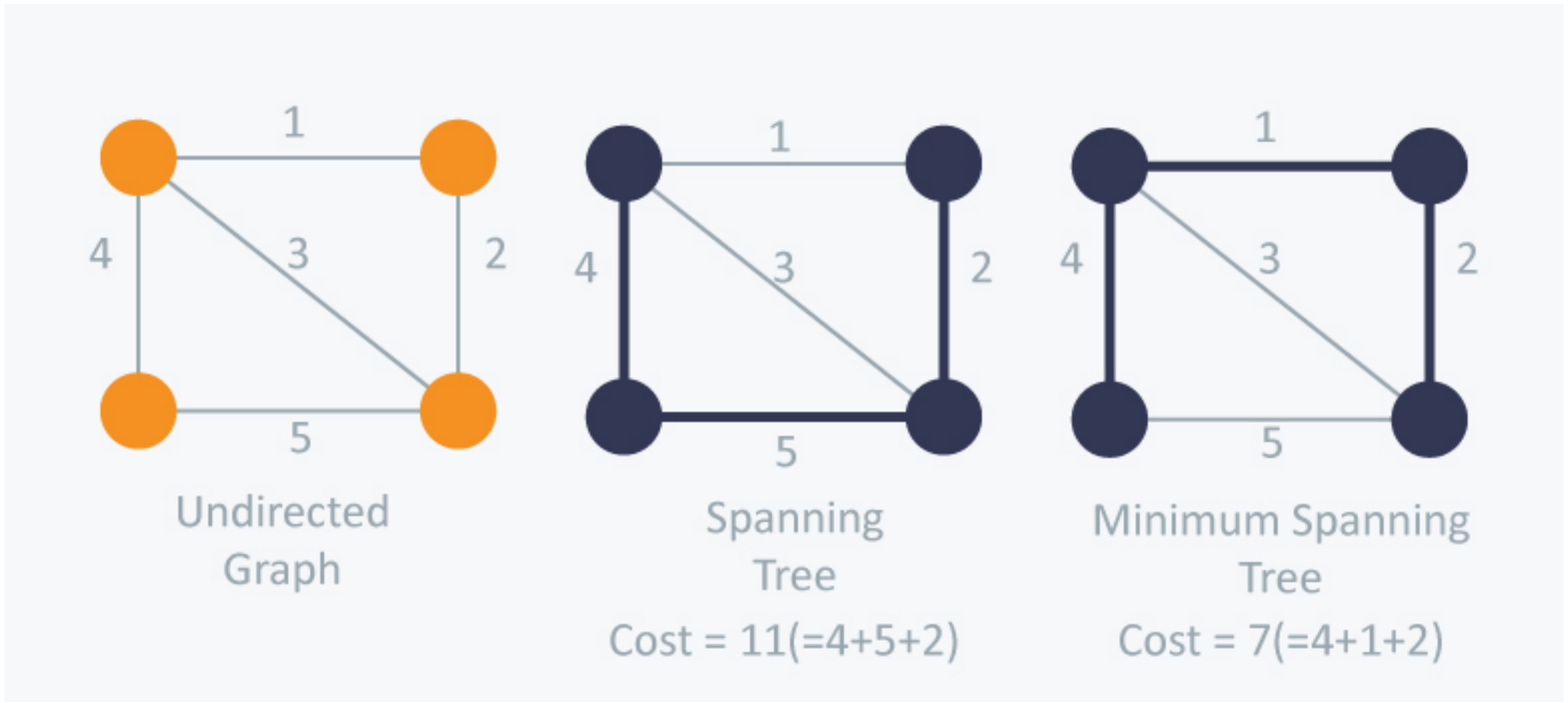


$M + N$ **VS** N^2

Minimum Spanning Tree

Spanning tree: for graph **G = (V, E)** is a subset of edges from **E**, forming a tree connecting all vertices of **V**.

For edge-waited graph we are usually interested in minimum value spanning tree



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Two main algorithms:

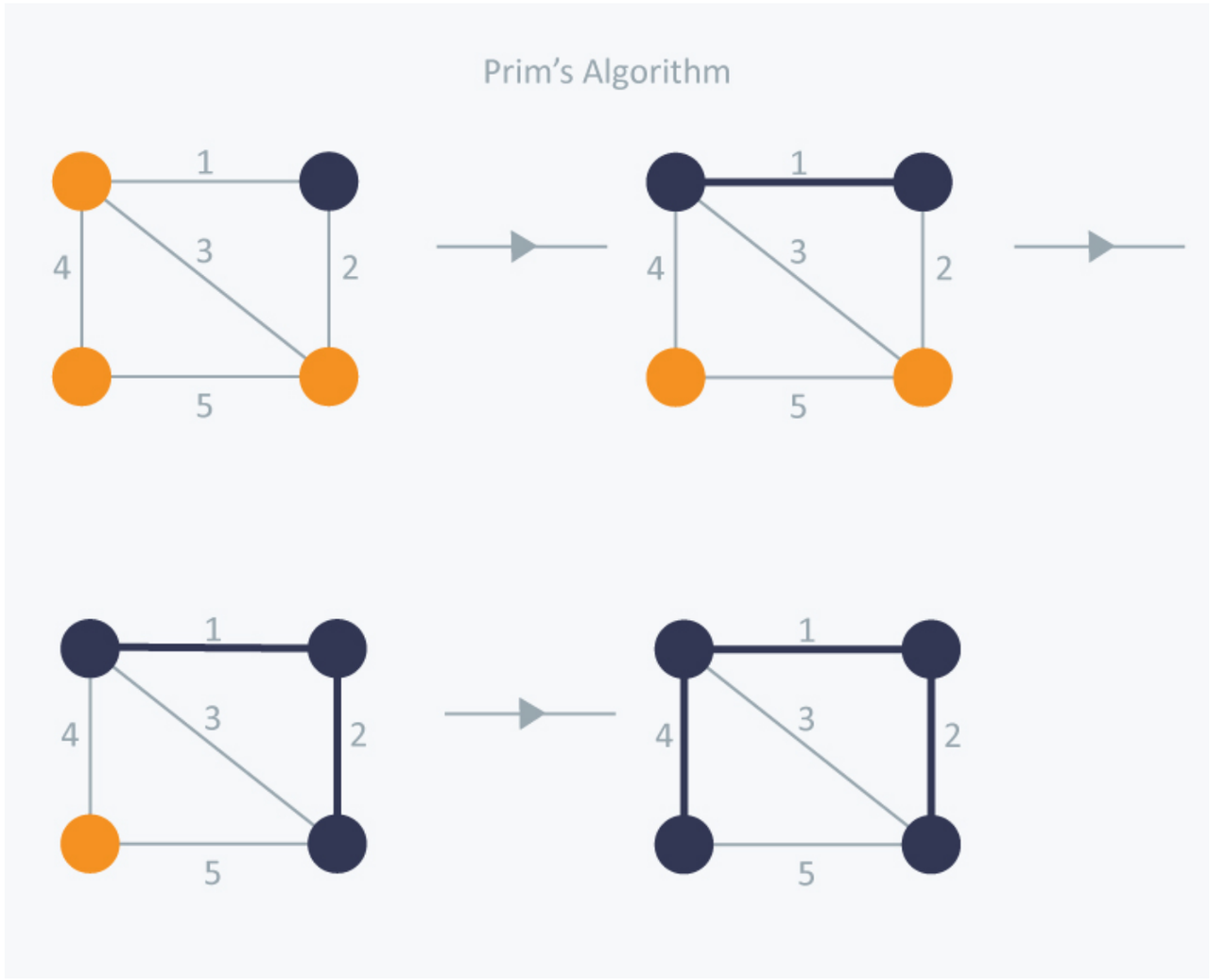
- Prim
- Kruskal

- Minimum spanning tree minimises total length (weight) over all possible spanning trees.
- There could be more than one minimum spanning tree in the graph.

Algorithms are greedy

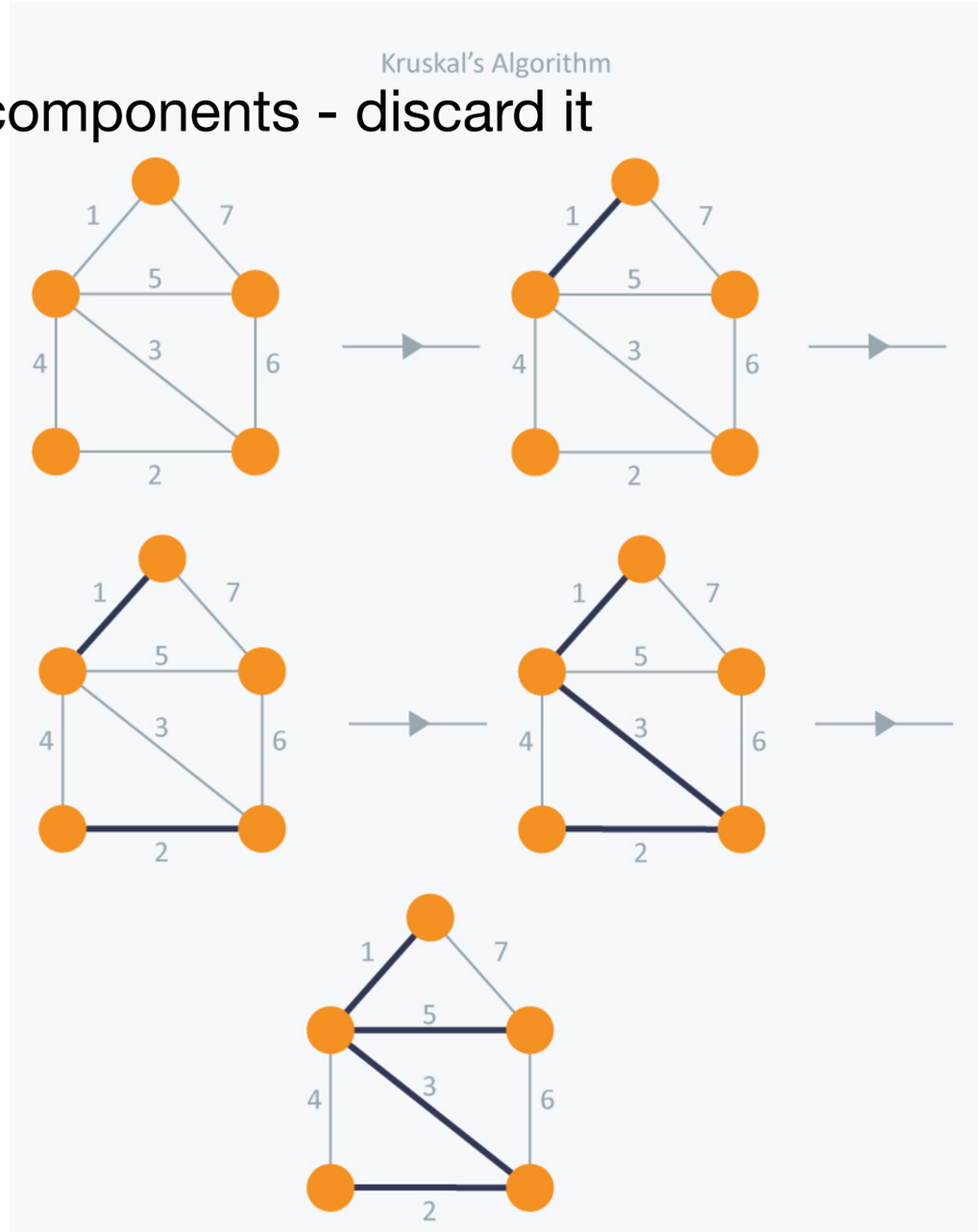
Prim algorithm

1. Select random node to start from
2. While non-tree nodes remaining:
    1. Select an edge of minimum weight between a tree and non-tree vertex
    2. Add selected edge and vertex to the tree



Kruskal algorithm

1. Initialise every node to be a single connected component
2. Consider the lightest edge
    1. If adjacent nodes are in the same connected components - discard it
    2. Otherwise, add edge, merge components



Prim vs Kruskal

- Prim **O(VE)**, if we don't keep track of cheapest edge
- Prim **O(VV)**, if we do, but use simple data structure
- Prim **O(E + V log V)**, if use priority queue
- Kurskal **O(E log E)**

- Prim **is better on dense graphs**
- Kruskal **is better on sparse graphs**

Shortest paths

Path is a sequence of edges connecting two vertices

- For unweighted graphs can be found with BFS
- Same for the graphs with equal weights

Dijkstra algorithm find shortest path between start and end vertices.

- O(NN)
- Greedy
- On each step selects the cheapest to add edge
- Almost Prim

Dijkstra algorithm

What is different from Prim:
Instead of considering only the weight of edge consider the length of the path from the start node